Prabhavit Gupta

Senior AI Engineer | Agentic Systems, AI-Native Dev (Claude Code/Codex) & Verifiable AI Governance

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Executive Summary

- **AI-Native Engineering & Agentic Systems:** Senior AI Engineer with 3 years commercial software experience leading AI-native development. Daily power-director of AI coding agents (Claude Code, Codex, Cursor), multiplying development velocity while maintaining uncompromising software engineering judgment, system security, and architecture integrity.
- **Verifiable, Governed AI Platforms & Tool Runtimes:** Deep specialization in agent runtimes, tool execution boundaries, and verifiable AI architectures. Architect of production multi-agent state machines with Model Context Protocol (MCP), enforcing Pydantic defensive schemas, deterministic human-in-the-loop gates, and zero-trust tool access.
- **Production Observability, Evals & System Design:** Extensive track record taking early-stage AI prototypes to robust, production-grade systems. Deep expertise in distributed trace observability with Langfuse, rigorous automated eval harnesses (DeepEval, RAGAS), and cloud infrastructure (Docker, AsyncIO, PostgreSQL/pgvector).

Technical Skills & Core Competencies

- **AI-Native Development:** Claude Code, Codex, Cursor, Automated Agentic Workflows, Spec-Driven Development, Context Window Management, Multi-Agent Pair Programming.
- **Agent Runtimes & Tooling:** Model Context Protocol (MCP), LangGraph, Autonomous Multi-Agent Loops, Tool Calling / Function Execution, Human-in-the-Loop Interrupts, State Persistence.
- **Core Software Engineering:** Python (AsyncIO, Typing, PyTest), FastAPI, TypeScript, Node.js, REST/GraphQL APIs, Docker, Git, CI/CD Automation, Defensive Security.
- **Data & Vector Infrastructure:** PostgreSQL (pgvector, HNSW indexing), Redis (State & Semantic Caching), SQL/NoSQL Architectures, Schema Migrations.
- **AI Governance & Observability:** Langfuse Distributed Tracing, DeepEval, RAGAS Benchmarks, Prompt Regression Harnesses, Hallucination Mitigations, Sandboxed Execution.

Professional Experience

Lead AI Engineer & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote Europe)

- Architected and deployed production-grade agentic platforms and verifiable AI systems for European tech companies, directing AI coding agents to deliver features at 5x individual engineering output.
- Engineered autonomous multi-agent state machines (LangGraph, Python AsyncIO) coordinating tool execution across databases, third-party APIs, and code sandboxes with strict deterministic boundaries.
- Built custom Model Context Protocol (MCP) servers and runtime connectors, ensuring granular permissioning, prompt injection defense, and structured Pydantic payload verification.
- Designed comprehensive CI/CD evaluation pipelines using DeepEval and RAGAS, preventing model drift, regression, and unauthorized tool calls before production releases.
- Integrated real-time distributed telemetry via Langfuse, profiling end-to-end trace latency, tool usage fidelity, token budgets, and automated fallback behaviors.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Engineered high-performance backend microservices in Python (FastAPI) and TypeScript supporting intelligent document parsing, structured entity extraction, and automated reasoning pipelines.
- Built resilient vector retrieval layers using PostgreSQL + pgvector and Redis, optimizing embedding caches to reduce API latency by 60% while guaranteeing high semantic accuracy.
- Established automated test suites (PyTest) and Git-based CI/CD workflows, raising engineering standards and system reliability across distributed development environments.
- Collaborated directly with founders and product teams to translate complex AI requirements into maintainable, production-ready system architectures.

Key Engineering Projects

- **Agentic Coding & Evaluation Testbed (Claude Code + MCP + Langfuse):** Built an autonomous multi-agent testbed orchestrating code generation, static analysis, unit test synthesis, and runtime execution with Langfuse distributed tracing and verification gates.
- **Verifiable Tool-Execution Gateway (Python + FastAPI + Docker):** Sandboxed execution environment

implementing Model Context Protocol (MCP) servers with cryptographic call auditing, Pydantic schema validation, and rate-limiting.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous training in formal logic, epistemology, decision theory, and complex mathematical systems analysis.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), Hindi (Native).
- **Work Authorization & Legal Status:** EU Long-Term Permanent Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted legal right to work across the EU without sponsorship; available immediately for remote full-time / B2B contract engagement.